

Statistical Validation Report

ApexPlanet Software Pvt. Ltd.

Internship Task 4 – Data Storytelling & Statistical Validation

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Executive Summary

This report validates key business insights derived from Exploratory Data Analysis (EDA) using statistical methods. The objective is to determine whether a significant relationship exists between customer order quantity and sales revenue.

Business Problem

Understanding the factors that influence sales performance is critical for improving business decisions.

The primary question addressed in this analysis is:

Does the quantity ordered significantly impact sales revenue?

Dataset Overview

The dataset contains historical sales transaction records including:

- Order Number
- Customer Name
- Product Line
- Quantity Ordered
- Price Each
- Sales Revenue
- Order Date Information

After data cleaning and preprocessing, the dataset was used for statistical analysis and business intelligence reporting.

Hypothesis Testing

Null Hypothesis (H_0)

There is no significant relationship between Quantity Ordered and Sales Revenue.

Alternative Hypothesis (H_1)

There is a significant relationship between Quantity Ordered and Sales Revenue.

Statistical Method

The Pearson Correlation Test was used to measure the strength and direction of the relationship between:

- Quantity Ordered
- Sales Revenue

Significance Level (α): 0.05

Results

The correlation analysis produced:

- Positive Correlation between Quantity Ordered and Sales Revenue
- P-value less than 0.05

This indicates a statistically significant relationship.

Interpretation

The analysis demonstrates that as the quantity ordered increases, sales revenue generally increases.

The observed relationship is not due to random chance and can be considered statistically significant.

Therefore:

Reject the Null Hypothesis (H_0)

Accept the Alternative Hypothesis (H_1).

Key Business Insights

Revenue Performance

- Total Revenue: \$10,032,628.85
- Average Sales: \$3,553.89
- Total Orders: 307
- Total Customers: 92

Product Performance

- Classic Cars generated the highest sales volume.
- Trains contributed the lowest sales volume.

Customer Analysis

- A small group of customers contributes a significant portion of total revenue.
- Customer retention strategies can improve long-term profitability.

Sales Behavior

- Larger order quantities are associated with higher revenue generation.
- Bulk purchase promotions may increase overall sales performance.

Business Recommendations

1. Focus marketing efforts on high-performing product categories.
2. Encourage bulk purchases through discounts and promotional campaigns.
3. Develop customer loyalty programs for top customers.
4. Monitor key performance indicators using interactive dashboards.
5. Continue statistical validation of future business strategies.

Conclusion

The statistical validation confirms a significant positive relationship between Quantity Ordered and Sales Revenue.

The findings support data-driven decision-making and provide actionable recommendations for improving revenue growth, customer engagement, and operational efficiency.

Prepared for ApexPlanet Software Pvt. Ltd. Internship Program

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